

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.7	12.1	1.6	0.3	0.1	53.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.4	11.4	1	0	0.1	55.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.4	11.7	1.9	0.3	0.1	53.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33	11.1	1.1	0	0.1	54.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.1	12.5	1.3	0.2	0.2	53.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	32	12.3	1.1	0.1	0.1	54.4
U2 - Efficacy additive alpha prior Beta(1, 1)	33.8	11.4	1.2	0.2	0.1	53.3
U3 - Efficacy additive alpha prior Beta(1, 1)	31.6	11.9	1.4	0.1	0.1	54.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	6.1	1.9	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.5	5.7	1.9	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15	6	2	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.4	5.7	1.9	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.1	6	2	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.5	5.6	1.9	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	15	6	2	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.5	5.6	1.9	0.9	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.3	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.2	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.4	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.2	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.4	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.2	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.4	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.2	0.6	0.4	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.7	21.8	37.8	11.8	1.9	1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.5	21	34.6	10.8	1.2	0.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.6	20.1	36.2	12.7	2.3	1.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32	21	34.7	10.3	1.2	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29	19.5	36.3	12.1	2.1	1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.9	21.2	34.3	8.3	1.3	1
U2 - Efficacy additive alpha prior Beta(1, 1)	28.9	20.2	36.3	11.4	2.3	0.9
U3 - Efficacy additive alpha prior Beta(1, 1)	33.8	21.4	33.3	9	1.3	1.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	7.7	8.8	3.1	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	7.7	8.5	2.9	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	7.4	8.6	3.3	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.5	7.7	8.4	2.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	7.3	8.7	3.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.6	7.6	8.4	2.9	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	10	7.3	8.6	3.3	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.6	7.5	8.4	2.9	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.4	1.7	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.4	1.6	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	1.7	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.4	1.6	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	1.7	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	1.6	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.4	1.7	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.4	1.6	0.8	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	49.7	25.7	10	1.6	2.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.5	54.8	20.9	7.9	1.2	2.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.2	50.9	24.4	10	1.7	2.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	56.7	20.2	7.7	1	2.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	49.6	25.4	9.5	1.6	2.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.1	54.7	21.7	6.6	1.2	2.7
U2 - Efficacy additive alpha prior Beta(1, 1)	11.9	48.9	25.6	9.2	1.6	2.8
U3 - Efficacy additive alpha prior Beta(1, 1)	12.7	54.2	22.3	6.8	1.2	2.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	12	6.7	3.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.4	12.7	6.1	2.9	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	11.8	6.5	3.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.5	12.6	6.2	2.9	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.3	11.4	6.6	3.4	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.6	12.2	6.2	2.9	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.4	11.4	6.6	3.4	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.7	12.3	6.2	2.9	0.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	4	1.8	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	4	1.6	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	4	1.7	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4	1.6	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	4	1.7	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4	1.6	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	4	1.7	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	4	1.6	0.9	0.2	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.2	6.9	19.1	53.9	13.6	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	7.8	19.7	51.3	11.3	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.1	7.4	19.5	53.4	13.4	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	7.9	19	52.1	11.1	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.5	8.2	19.8	51.6	13.7	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	8.1	19.5	51.2	11.1	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	6.4	8.4	20.2	51.2	13.6	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	10.4	8.7	19.2	50.9	10.6	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.3	4.8	7.1	10.6	3.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	4.9	7.1	10.5	2.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	4.8	7.1	10.4	3.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	4.9	7.1	10.5	2.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	4.8	7.1	10.2	3.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	4.8	7.2	10.3	2.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	4.8	7.1	10.1	3.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	4.8	7.2	10.3	3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.6	2.5	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.6	2.4	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.1	1.6	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.6	2.4	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3	1.6	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.6	2.4	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.1	1.6	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.1	1.6	2.4	0.6	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	35.8	24.9	24.2	9.7	0.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.4	44.6	22.5	19.1	6	0.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	34.7	26.5	24	9.2	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.8	43.5	23.3	18.6	6.4	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	34.7	27.4	21.1	11	0.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.8	43	22.7	19.3	6.8	0.4
U2 - Efficacy additive alpha prior Beta(1, 1)	5.3	33.9	26.2	23.3	10.9	0.4
U3 - Efficacy additive alpha prior Beta(1, 1)	7.4	42.4	22.8	20	7.1	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	7.6	7.2	7.2	4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.2	8.5	7.3	6.8	3.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	7.4	7.1	7.2	4.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	8.2	7.3	6.8	3.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	7.1	7.2	7.1	4.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	8	7.2	6.9	3.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.1	7.2	7.1	7.2	4.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.2	8	7.3	6.9	3.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	3.3	2	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.3	2	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	3.3	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.3	2	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.2	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.4	1.9	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.4	1.9	2.1	0.9	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	65.9	15.8	3.6	1	0.2	13.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.8	15.3	3.6	0.7	0.2	13.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	66.3	14.9	4.1	0.9	0.2	13.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.9	13.9	3.5	0.7	0.2	13.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	67.2	13.7	4.3	0.9	0.2	13.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.9	13.7	3.2	0.6	0.2	13.4
U2 - Efficacy additive alpha prior Beta(1, 1)	68.2	12.9	4.4	0.9	0.2	13.4
U3 - Efficacy additive alpha prior Beta(1, 1)	70.3	12.7	2.9	0.4	0.2	13.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.4	6	2.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.8	6	1.9	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.5	5.9	2.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.1	5.7	1.9	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.5	5.7	2.2	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.2	5.6	1.9	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.6	5.7	2.2	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.3	5.5	1.9	0.8	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.6	0.6	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.5	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.6	0.6	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.5	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.5	0.6	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.5	0.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.5	0.6	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.5	0.5	0.2	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.3	50.7	24.8	8.7	2.3	1.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.4	56.3	20.7	6.5	1.1	1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13	49.7	24.7	9	2.6	1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.9	54.4	21.2	6.2	1.2	1.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.9	47.3	24	10.1	2.6	1.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.7	52.7	21.8	6.6	1.2	1
U2 - Efficacy additive alpha prior Beta(1, 1)	15.6	48.3	23.2	9.5	2.4	1
U3 - Efficacy additive alpha prior Beta(1, 1)	17.2	52.4	21.7	6.5	1.1	1.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8	13.4	5.8	2.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.3	14.4	5	1.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	13	5.8	2.3	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	13.9	5.1	1.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.2	12.5	5.9	2.6	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	13.3	5.3	2	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.2	12.5	5.9	2.5	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.8	13.4	5.3	2	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.6	1.3	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.1	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.6	1.3	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.6	1.3	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.5	1.1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.6	1.3	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.6	1.1	0.5	0.1	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	44.8	12.3	1.6	0.7	0.3	40.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	45.1	11.7	1.7	0.4	0.2	40.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	45.4	11.7	2.3	0.5	0.2	39.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	45.9	11	1.5	0.3	0.1	41.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	45.5	10.5	2.6	0.5	0.2	40.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	46.1	10.8	1.2	0.4	0.1	41.4
U2 - Efficacy additive alpha prior Beta(1, 1)	45.6	10.9	2.3	0.7	0.2	40.3
U3 - Efficacy additive alpha prior Beta(1, 1)	46.3	10.8	1.4	0.4	0.1	41
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	5.8	1.8	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.7	5.6	1.7	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.5	5.8	1.8	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.8	5.5	1.6	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.5	5.7	1.8	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.8	5.5	1.7	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.5	5.7	1.8	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.8	5.5	1.7	0.7	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	2.9	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	2.8	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	2.9	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	2.8	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	2.9	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	2.8	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	2.9	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	2.8	0.5	0.3	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	16.6	23	17.5	21.7	5.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22	20.7	18.6	15	18.3	5.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.1	18.1	20.7	17.9	22.8	5.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.6	18.3	19.9	14.3	18.6	5.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.6	16	22.3	17.1	21.8	5.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.6	18.9	19.3	14.9	18.9	5.4
U2 - Efficacy additive alpha prior Beta(1, 1)	17.7	16.1	21.6	16.6	22.6	5.4
U3 - Efficacy additive alpha prior Beta(1, 1)	23.8	18.1	19	15.4	18.6	5.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.6	6.4	6.1	5.1	4.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.2	6.3	5.9	5.2	3.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.7	6.3	6	5.1	4.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.4	6.2	5.8	5	4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.8	6.3	6	5	4.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.5	6.2	5.8	5	4	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.8	6.3	5.9	5	4.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.5	6.2	5.8	5	4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.8	1.6	1.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.8	1.6	1.7	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.8	1.6	1.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.7	1.5	1.7	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.8	1.6	1.8	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.8	1.5	1.7	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.8	1.6	1.8	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.8	1.6	1.7	0.8	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14	41.2	15.1	4.2	0.9	24.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	41.4	13	3.7	0.8	25.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.3	41.1	14.4	4.6	1	24.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.5	42.8	12.5	3.4	0.8	25
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.8	40.5	15.2	4.3	0.9	24.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16	42.9	12.1	3.6	0.7	24.7
U2 - Efficacy additive alpha prior Beta(1, 1)	14.2	40.8	14.8	4	1	25.2
U3 - Efficacy additive alpha prior Beta(1, 1)	15	44.1	12.3	3.4	0.7	24.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	10.5	4.3	1.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	10.9	3.9	1.5	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	10.4	4.2	1.8	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	10.8	4	1.5	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	10.2	4.3	1.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	10.7	4	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.8	10.3	4.3	1.8	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	11	10.7	4	1.6	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4.2	1.1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.2	1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4.2	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.1	1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.2	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.2	0.9	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	4.2	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.2	1	0.5	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.3	23.3	4.2	0.1	0	6.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	68.2	22.6	2.7	0.2	0	6.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	66.8	22.6	4	0.1	0	6.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70	21.5	2.2	0.2	0	6.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	67.1	21.9	4.5	0.1	0	6.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	72.3	19	2.5	0	0	6.2
U2 - Efficacy additive alpha prior Beta(1, 1)	68	21.3	4.2	0.2	0	6.3
U3 - Efficacy additive alpha prior Beta(1, 1)	72.7	18.5	2.5	0.1	0	6.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	7.7	2.9	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18	7.6	2.5	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.5	7.7	2.8	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.3	7.4	2.5	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.6	7.5	2.9	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.4	7.2	2.6	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.7	7.5	2.8	1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.3	7.2	2.6	0.9	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.6	0.9	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.5	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.5	0.9	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.4	0.8	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.5	0.9	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.4	0.8	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.5	0.9	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.4	0.8	0.4	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	12.1	12.2	44.6	19	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.1	15.4	9.1	40.7	17.4	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	12.8	11.8	45.3	18.8	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.2	13.5	8.5	42	16.5	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	12.6	12.6	46.3	17.3	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.1	14.6	8.7	43	16.3	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	11	12.4	11.5	46.7	18.1	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	17.6	13.6	8.7	43.5	16.3	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	5.5	5.7	9.3	4.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	5.6	5.6	9.3	4.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	5.4	5.8	9.2	4.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	5.5	5.6	9.3	4.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	5.4	5.7	9.1	4.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	5.5	5.6	9.2	4.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.9	5.4	5.7	9.1	4.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.2	5.5	5.6	9.2	4.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.2	1.6	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.2	1.5	2.3	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.2	1.6	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.2	1.5	2.3	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.3	1.6	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.3	1.5	2.3	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.3	1.6	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.3	1.5	2.3	1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.4	40.6	14.9	5	1	21.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.6	41.6	12.8	3.9	0.9	22.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18	39.6	14.8	4.9	0.8	21.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.2	43	12.6	3.6	0.9	21.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.1	40.8	15.4	3.9	1	21.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.3	40.5	13.7	3.9	0.7	21.9
U2 - Efficacy additive alpha prior Beta(1, 1)	17	40.3	15.6	4.3	0.8	22
U3 - Efficacy additive alpha prior Beta(1, 1)	19.6	41.1	13.3	3.7	0.7	21.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	9.7	4.6	2.3	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11	10	4.3	2.1	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	9.6	4.5	2.4	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	9.8	4.4	2.2	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	9.6	4.6	2.4	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	9.7	4.4	2.2	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.7	9.6	4.6	2.4	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.1	9.7	4.4	2.2	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.4	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.4	1.1	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.4	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.4	1.1	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.4	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.3	1.1	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.4	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.4	1.2	0.7	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.2	18.4	16.8	32.4	12.3	0.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.6	22.9	11.5	28.4	8.7	0.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.8	19.8	16.2	32.5	11.6	1.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.1	22	12.9	27.5	9.6	0.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.1	18.6	16.2	33.3	11.6	1.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.3	20.8	13.9	27.9	10	1.1
U2 - Efficacy additive alpha prior Beta(1, 1)	17.9	18.9	18	32.1	12	1.1
U3 - Efficacy additive alpha prior Beta(1, 1)	26.7	20.5	13.7	28	10.2	0.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.4	6.6	5.9	7.2	3.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	6.5	5.6	7.2	3.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.5	6.5	5.9	7.2	3.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	6.4	5.6	7.2	3.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.5	6.5	5.8	7.2	3.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.3	6.4	5.6	7.1	3.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.6	6.4	5.8	7.2	3.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.3	6.4	5.6	7.1	3.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.7	1.7	2.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	1.7	2.1	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.7	1.7	2.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.7	1.6	2	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.7	1.7	2.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.7	1.6	2.1	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.7	1.7	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.7	1.6	2.1	0.8	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	48.2	32.3	12.5	4.4	0.3	2.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	54.8	30.4	9.4	3.1	0.1	2.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	48.8	31.5	12.7	4.4	0.4	2.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	57.9	27.6	9	3.1	0.2	2.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	47.4	32.1	13	5	0.3	2.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	59.1	27	8.5	3	0.2	2.2
U2 - Efficacy additive alpha prior Beta(1, 1)	47.1	32.4	13.2	4.8	0.4	2.1
U3 - Efficacy additive alpha prior Beta(1, 1)	59.4	26.6	8.5	3	0.2	2.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	8.7	5.7	3.7	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	8.4	5.4	3.5	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.6	8.6	5.7	3.6	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	8.3	5.3	3.5	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.7	8.6	5.7	3.6	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.8	8.2	5.3	3.4	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.8	8.5	5.7	3.7	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.8	8.2	5.3	3.4	1.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4.2	1.8	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	4.1	1.7	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.2	1.8	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	4.2	1.7	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4.2	1.8	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	4.1	1.7	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.2	1.8	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.3	4.1	1.7	1.4	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	38.1	26.9	15.2	12.2	7.6	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	44	27.4	14.7	9.8	4.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	38.2	26.4	15.1	12.5	7.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	42.9	28.6	14.4	10	4.1	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	37.3	26.9	15.8	11.6	8.4	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	42.8	29.1	14.9	8.7	4.5	0
U2 - Efficacy additive alpha prior Beta(1, 1)	37.6	26.9	15.7	11.7	8.1	0
U3 - Efficacy additive alpha prior Beta(1, 1)	44.1	27.9	13.6	9.7	4.7	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.3	4.9	5.9	8.3	6.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	5.3	6.2	8.2	5.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	4.8	5.9	8.3	6.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	5.2	6.2	8	5.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.3	4.7	6.1	8.1	6.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	5.1	6.3	8	5.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	4.7	6	8	6.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.9	5	6.3	8	5.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	2.1	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2.2	2.8	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	2	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	2.2	2.8	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	2	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	2.1	2.9	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	2	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	2.1	2.9	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.7	62.6	17	1.5	0	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.1	64.7	12.4	0.7	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.8	60.2	17.3	1.6	0	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25	61.2	12.8	0.9	0	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.6	58.8	17	1.4	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.4	59.5	13	1	0	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	22.9	58.3	17	1.5	0.1	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	26.1	60.2	12.7	0.9	0	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	15	4.6	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	15.5	3.9	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	14.6	4.8	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	14.9	4	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	13.9	5.1	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.7	14.1	4.3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.9	13.7	5.1	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.8	14.3	4.1	0.8	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.9	1.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.9	1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.9	1.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.9	1	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.8	1.2	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.8	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.8	1.2	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.8	1	0.3	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	11.4	8.5	16.7	53.7	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17	14	9.7	13.6	45.7	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.5	11	9.9	16.6	54	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.6	13.4	9.8	14.3	45.9	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	10.3	9.6	16.9	54.4	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	13.6	8.8	14.7	48.3	0
U2 - Efficacy additive alpha prior Beta(1, 1)	7.9	10.8	10.3	16.3	54.7	0
U3 - Efficacy additive alpha prior Beta(1, 1)	14.2	12.5	9.7	15	48.6	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4.1	5.2	7	10.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	4.2	5.4	7.1	9.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	4	5.1	6.9	10.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.7	4.2	5.3	7	9.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.9	5.1	6.9	10.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	4.2	5.2	6.9	10	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	4	5.1	6.8	10.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.8	4.1	5.2	6.8	10.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.5	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.5	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.5	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.5	2.5	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.5	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.5	2.5	1.5	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.6	59.4	19.4	1.4	0	3.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20	59.6	15.9	1.3	0	3.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.3	60.6	18.2	1.5	0	3.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.5	60.5	14.5	1.5	0	3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.2	59.5	17.3	1.8	0	3.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.8	58.7	15.2	1.3	0.1	2.9
U2 - Efficacy additive alpha prior Beta(1, 1)	18.1	61.2	16.1	1.3	0	3.3
U3 - Efficacy additive alpha prior Beta(1, 1)	20.4	59.7	15.3	1.6	0	3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	13	6.2	1.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	13.2	5.9	1.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	13	6.1	1.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9	13.1	5.8	1.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	12.7	6.2	2	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.2	12.9	5.7	1.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.7	12.8	6.1	2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.1	12.9	5.8	1.8	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.8	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.7	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.8	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.7	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.8	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.8	1.5	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.9	1.6	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.8	1.5	0.7	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	50.3	29.1	10.3	0.8	0.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.9	53.2	23.1	8.7	0.7	0.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	48.9	29.5	10.9	0.9	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.1	54.7	22	8.1	0.7	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	50.1	27.3	10.1	0.9	0.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.1	54.4	21.4	8	0.7	0.4
U2 - Efficacy additive alpha prior Beta(1, 1)	10.6	49.6	28.7	9.6	1.1	0.4
U3 - Efficacy additive alpha prior Beta(1, 1)	15	53.4	22.4	7.9	0.9	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	10.6	8.3	4.7	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	11	7.8	4.4	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	10.5	8.3	4.6	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	11	7.8	4.4	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	10.2	8.2	4.8	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.6	10.8	7.9	4.4	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.2	10.1	8.3	4.8	1.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.6	10.7	7.8	4.5	1.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.9	2.3	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4	2.2	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4	2.3	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4	2.2	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4	2.2	1.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.1	2.2	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	4	2.2	1.7	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	4	2.2	1.5	0.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	55.8	1.4	0	0	0	42.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	56.5	1.2	0	0	0	42.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	56.8	0.9	0	0	0	42.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	56.5	1.3	0	0	0	42.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	56.8	0.9	0	0	0	42.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	56.2	1	0	0	0	42.8
U2 - Efficacy additive alpha prior Beta(1, 1)	56.9	1	0	0	0	42.1
U3 - Efficacy additive alpha prior Beta(1, 1)	57.2	0.8	0	0	0	42
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.1	2.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.2	2.4	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.2	2.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.3	2.3	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	2.3	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.3	2.3	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20.2	2.3	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.4	2.3	0.1	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.1	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.5	64.1	12.7	1.8	0	1.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.2	62	11.8	1.8	0	2.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.5	63.9	12.5	2.1	0	2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.5	61	11	2.4	0	2.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.2	63	11.4	2.4	0	2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.4	58.6	11.5	2.6	0	1.9
U2 - Efficacy additive alpha prior Beta(1, 1)	20.8	62.7	11.9	2.7	0	1.9
U3 - Efficacy additive alpha prior Beta(1, 1)	26	57.3	11.7	2.8	0	2.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9	13.5	5.4	1.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	13.5	5	1.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9	13.5	5.4	1.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	13.4	5	1.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.2	13.2	5.4	1.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	13.2	5.1	1.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.2	13.3	5.3	1.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.8	13.1	5.1	1.8	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.9	1.7	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.7	1.6	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.9	1.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.9	1.5	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.9	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	4.9	1.6	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	4.9	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	4.8	1.6	0.6	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.5	2.9	0	0	0	78.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.7	3.1	0	0	0	78.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.9	2.8	0	0	0	78.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19	2.4	0	0	0	78.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19	3.2	0	0	0	77.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.2	2.6	0	0	0	78.2
U2 - Efficacy additive alpha prior Beta(1, 1)	18.6	3.2	0	0	0	78.2
U3 - Efficacy additive alpha prior Beta(1, 1)	19.1	2.6	0	0	0	78.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	4.4	0.5	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.8	4.2	0.5	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.9	4.4	0.5	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.9	4.2	0.5	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	4.3	0.5	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16	4.2	0.5	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.9	4.4	0.5	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.9	4.2	0.5	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.6	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.6	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.6	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.6	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.5	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.1	0.6	3.2	56.3	37.8	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.8	2.5	4	58.9	31.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.7	0.9	3.6	55.4	38.4	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	2.6	2.2	4.4	58.3	32.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.5	0.9	4.9	54.9	37.8	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3	1.7	5.1	58	32.2	0
U2 - Efficacy additive alpha prior Beta(1, 1)	1.3	1	4.8	55.7	37.2	0
U3 - Efficacy additive alpha prior Beta(1, 1)	2.5	1.8	6	58.2	31.5	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	3.9	12.8	7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	4	13.4	6.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.1	3.9	12.7	7.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.2	4	13.2	6.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.1	3.9	12.4	7.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	4	12.9	6.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.1	3.9	12.4	7.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.2	4.1	12.9	6.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.5	1.1	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1.1	2.4	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.5	1.1	2.5	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.5	1.1	2.5	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1.1	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0	2.5	1.1	2.5	1.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.4	33.6	34.4	4.2	0.1	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.4	33.9	29.4	3.2	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.8	33.8	33.2	4	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	35	33.7	28.1	3.1	0	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.7	33.2	33	4	0	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	36	33.2	27.9	2.8	0	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	29	33.7	33.5	3.6	0.1	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	36.9	30.8	28.9	3.2	0.1	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9	9.9	8.4	2.5	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	9.9	8.2	2.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.1	9.6	8.5	2.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	9.7	8.1	2.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	9.3	8.6	2.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	9.4	8.1	2.3	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.2	9.4	8.6	2.5	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.1	9.4	8	2.3	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.7	1.9	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.7	2	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.7	1.8	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.7	1.9	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.7	1.8	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.8	1.9	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.7	1.8	0.9	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.4	35.9	31.3	5.4	0.1	3.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.4	35.8	27.5	4.8	0.1	3.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.2	35.2	30.7	4.2	0.1	3.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.2	34.8	27.6	3.8	0.1	3.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.9	33.8	31.7	3.9	0.1	3.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.2	34.7	27.1	3.5	0.1	3.4
U2 - Efficacy additive alpha prior Beta(1, 1)	26.7	34	31.5	3.9	0.1	3.8
U3 - Efficacy additive alpha prior Beta(1, 1)	31.8	32.8	28.3	3.5	0.1	3.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	9.5	8	2.7	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	9.5	7.6	2.5	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	9.3	7.8	2.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	9.4	7.5	2.5	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	9.1	7.8	2.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	9.2	7.5	2.5	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	9.1	7.9	2.7	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.2	9.2	7.5	2.5	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.2	1.7	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.2	1.5	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.2	1.6	1.2	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.2	1.5	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.2	1.6	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.2	1.5	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.2	1.6	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.1	1.5	1.1	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.8	25.9	46.9	9.9	0.4	1.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.7	28.3	42	7.6	0.3	1.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.7	27.4	45.3	9.2	0.4	1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.4	27.3	41.6	7.2	0.4	1.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.7	26.4	46.7	7.9	0.3	1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.8	26.7	43	7.4	0	1.1
U2 - Efficacy additive alpha prior Beta(1, 1)	18.2	25.6	47.3	7.7	0.3	0.9
U3 - Efficacy additive alpha prior Beta(1, 1)	23.1	25.6	43.9	6.3	0	1.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.9	8.7	10.2	3.6	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.4	8.8	9.8	3.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7	8.6	10	3.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.6	8.6	9.8	3.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.1	8.4	10	3.7	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.7	8.5	9.9	3.3	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.1	8.4	10.1	3.6	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.8	8.5	9.8	3.3	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.8	2.2	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.8	2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.8	2.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.8	2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.8	2.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.9	2	1.2	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.8	2.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.8	2	1.2	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.9	47.5	6.5	0	0	23.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.5	46.7	5.9	0.1	0	23.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.4	47.1	7	0.1	0	23.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.6	46.6	5.7	0.1	0	24
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.6	49.6	5.5	0.1	0	23.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.5	46.6	5.5	0.2	0	23.2
U2 - Efficacy additive alpha prior Beta(1, 1)	23	48	5.7	0.1	0	23.2
U3 - Efficacy additive alpha prior Beta(1, 1)	24.2	46	5.5	0.2	0	24.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.1	11.6	3.4	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.3	11.6	3.2	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.1	11.5	3.5	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.4	11.4	3.3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.2	11.4	3.5	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.5	11.3	3.3	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.1	11.4	3.5	1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.5	11.4	3.2	0.8	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.6	1.1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.6	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.6	1.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.6	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.7	1.1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.7	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.7	1.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.7	1	0.3	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	62.2	30.7	4.6	0.1	0	2.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	65.3	30.1	2.3	0.1	0	2.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	63.3	30.7	3.7	0.1	0	2.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	68.4	26.9	2.1	0.2	0	2.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	67.2	27.4	3.1	0.1	0	2.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	71.3	24.2	2	0.1	0	2.4
U2 - Efficacy additive alpha prior Beta(1, 1)	67.4	27.1	3	0.2	0	2.3
U3 - Efficacy additive alpha prior Beta(1, 1)	70.9	24.2	2.3	0.3	0	2.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.9	9.1	2.2	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.8	8.7	1.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.2	8.6	2.3	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.3	8.2	1.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.3	8.3	2.4	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.6	7.8	1.8	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.5	8.2	2.4	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.6	7.8	1.8	0.4	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.2	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.1	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.1	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.9	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.9	0.6	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.3	11.8	14.7	44	12.2	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.2	15	12.9	37.1	10.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.3	11.6	14.4	44.8	11.9	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25	15.2	11.7	37.7	10.4	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.2	12.5	15.6	43	11.7	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.3	15.2	12.6	38.7	9.2	0
U2 - Efficacy additive alpha prior Beta(1, 1)	16.3	12.2	16.1	43.5	11.9	0
U3 - Efficacy additive alpha prior Beta(1, 1)	25.2	14.1	12.4	39.2	9.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	5.1	6.4	10.1	4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	5.3	6.4	10	3.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	5.1	6.4	10	4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	5.3	6.3	9.9	3.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	4.9	6.4	9.9	4.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	5.2	6.3	9.8	3.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.6	4.9	6.3	9.9	4.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5	5.2	6.3	9.9	3.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	1.8	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	1.8	2.4	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	1.8	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3	1.7	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	1.7	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	1.7	2.3	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	1.8	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.7	2.3	1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.7	36.4	30.5	23.5	5.9	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.7	41.3	29.5	20.5	3	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	36.9	28.6	23.3	6.2	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.6	41.7	29.2	19.7	2.8	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	34.4	29.7	24	6	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.9	40.3	30	18.6	3.2	0
U2 - Efficacy additive alpha prior Beta(1, 1)	5.9	35.2	29	23.7	6.2	0
U3 - Efficacy additive alpha prior Beta(1, 1)	8.2	40.6	28.9	19	3.3	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	12.1	7.6	5	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	12.9	7.5	4.6	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	11.7	7.5	5.1	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	12.6	7.4	4.7	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	10.8	7.8	5.5	1.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	11.8	7.7	4.9	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	4	10.8	7.7	5.6	2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.2	11.9	7.5	4.8	1.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.6	1.1	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	1.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.5	1	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.5	1.2	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.5	1.1	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	1.2	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.5	1	0.4	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.7	14.8	33.1	28.8	11.6	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	16.7	31.7	27.5	6.6	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	13.9	35.3	27.8	11.2	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.3	17.3	33.3	24.3	6.8	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	12.4	36.7	27.8	11.2	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.3	17.2	34.2	24.4	7.9	0
U2 - Efficacy additive alpha prior Beta(1, 1)	11.2	13.1	37.9	26.3	11.5	0
U3 - Efficacy additive alpha prior Beta(1, 1)	16.7	16.9	35	22.9	8.5	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4.3	8.6	8.7	4.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	4.6	9.1	8.6	4.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	4.2	8.6	8.6	5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.7	4.5	9	8.5	4.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	4.1	8.7	8.4	5.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.7	4.4	9	8.4	4.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	4.2	8.6	8.5	5.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.8	4.4	9	8.3	4.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.9	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.9	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.9	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.8	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.8	2.7	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.8	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.9	2.7	1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.4	5.4	11.5	46.8	27.9	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.6	7.7	12.9	44	22.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.4	6.1	11	46.4	28.1	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	8.6	12.4	45.1	22.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.5	6.4	12.3	45.3	28.5	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	8.8	13.1	44.2	22.3	0
U2 - Efficacy additive alpha prior Beta(1, 1)	7.8	6.3	11.6	46.2	28.1	0
U3 - Efficacy additive alpha prior Beta(1, 1)	11.6	8.6	13	42.7	24.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.7	6.1	10.6	6.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.9	6.2	10.7	5.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.4	3.7	6	10.4	6.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	3.8	6.1	10.8	5.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.4	3.7	5.9	10.1	6.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	3.8	6.2	10.5	5.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.4	3.7	5.9	10.2	6.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.8	6.2	10.4	6.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.7	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.7	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.7	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.6	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.8	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.7	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.2	15.4	46.7	20.5	3	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.5	16.4	42.4	20.1	2.3	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.5	15.4	49	18.6	3.3	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.8	16.5	42.8	18.4	2.3	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.6	15.5	48.7	18.1	3.8	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.6	16.7	43.5	17.6	2.4	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	13.2	15.7	49.1	18	3.8	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	20	15.4	44.1	17.9	2.4	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	6.8	9.8	5.9	2.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6	6.7	9.7	5.7	1.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	6.6	9.9	5.8	2.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.1	6.7	9.7	5.6	1.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	6.6	9.8	5.7	2.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.1	6.5	9.8	5.6	1.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.6	6.5	9.8	5.8	2.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.2	6.5	9.8	5.5	2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	2.1	1.9	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.6	2.1	1.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.5	2.2	1.9	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.6	2.1	1.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	2.1	1.9	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.6	2	1.8	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.5	2.1	1.9	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.6	2	1.8	0.5	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	41.1	33.6	13.4	2.3	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.7	46.7	28.4	10	1.2	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	39.5	33.1	13.4	3	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.7	47.6	28.2	9.2	1.3	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.6	40.3	30.2	14	2.9	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.4	45	28.4	9.7	1.5	0
U2 - Efficacy additive alpha prior Beta(1, 1)	12.3	40.1	30.1	14.6	2.9	0
U3 - Efficacy additive alpha prior Beta(1, 1)	15.5	45.9	27.6	9.7	1.3	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	12.1	8.5	3.9	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	12.9	8.1	3.6	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	11.7	8.4	4.1	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	12.8	7.9	3.5	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	11	8.3	4.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	12	8.2	3.7	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.7	11	8.4	4.4	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	5	12.2	8	3.7	1.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.1	2	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.1	1.9	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.1	1.9	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.8	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.1	1.8	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.8	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.1	1.9	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.1	1.8	1.1	0.2	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.3	46.9	17.7	4.5	1.2	8.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26	47.6	13.2	4.3	0.7	8.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.1	46.7	16.1	5.2	1.3	8.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.5	46.7	12.3	5.2	0.6	8.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.9	47.9	15.7	5	1.3	8.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.2	46.4	11.7	5.1	1	8.6
U2 - Efficacy additive alpha prior Beta(1, 1)	21.6	47.8	15.3	5.6	1.4	8.3
U3 - Efficacy additive alpha prior Beta(1, 1)	26.8	46.1	11.6	5.7	1	8.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	10.9	5.4	2.9	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	10.7	5.1	2.8	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	10.7	5.3	3	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	10.7	5.1	2.8	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	10.6	5.3	3	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.1	10.6	5.1	2.8	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.5	10.7	5.4	3	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.1	10.6	5.1	2.9	0.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.9	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.9	1.5	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.9	1.5	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.9	1.5	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	5	1.5	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.9	1.5	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	5	1.5	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.9	1.5	1	0.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.3	58.3	14.6	2.6	0.3	3.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.2	57.8	10.3	2.3	0.2	4.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.8	57.5	14.2	3	0.3	4.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.7	56.3	11.3	2.3	0.3	4.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.2	57	14.8	2.8	0.2	4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.2	54.1	12	2.4	0.3	4
U2 - Efficacy additive alpha prior Beta(1, 1)	21.4	56.7	14.9	2.8	0.2	4
U3 - Efficacy additive alpha prior Beta(1, 1)	27.1	53.7	12.3	2.3	0.2	4.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	12.2	5.4	2.6	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	12	5.2	2.4	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.1	12	5.4	2.6	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	11.9	5.2	2.4	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.1	11.8	5.5	2.6	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	11.7	5.1	2.4	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.1	11.9	5.5	2.6	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.8	11.8	5.2	2.4	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	5.2	1.6	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	5.1	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	5.2	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	5.1	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	5.2	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	5.2	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	5.3	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	5.2	1.5	0.9	0.2	